S.I.R.E.N.: Sensor Fusion and Multi-Layered Architecture for Emergency Preemption Systems

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Abstract—Emergency medical transit in densely populated urban networks is severely bottlenecked by traffic congestion, inadequate preemption systems, and uncoordinated infrastructure, leading to catastrophic delays during the critical "Golden Hour." This paper proposes S.I.R.E.N. (Smart Intelligent Response and Emergency Navigation), a novel 7-layer architecture designed to resolve these critical transit blockages. Unlike conventional systems that rely on prohibitive single-modality sensors, S.I.R.E.N. utilizes a trilateral sensor fusion approach combining macro-level cloud GPS routing, deterministic sub-GHz V2X micro-actuation, and edge-AI visual-acoustic fallbacks. The architecture spans seven layers—from the client interfaces and IoT hardware edge nodes to a cloud backend, event broker, geospatial database, and an integrated RoIP alerting gateway. By addressing critical failure modes such as queue spillback and secondary gridlock, the system ensures dynamic cross-traffic rebalancing and seamless preemptive traffic signal control. Consequently, S.I.R.E.N. aims to drastically reduce emergency vehicle travel times to align with global benchmarks. In the state of Odisha alone, which recorded 6,142 road accident fatalities in 2024, deploying this architecture targets a preventable mortality pool of approximately 1,754 deaths annually. Through mitigating transit delays, this sensor-fused multi-layered architecture provides a scalable, cost-effective framework to preserve the physiological temporal gradient of acute patients and significantly reduce pre-hospital mortality rates.

***Index Terms*—Edge Computing, Emergency Preemption, Geospatial Routing, Sensor Fusion, V2X Communication.**

I. INTRODUCTION

The physiological decline of trauma and acute medical patients follows a strict temporal gradient, necessitating clinical adherence to the "Golden Hour" [1]. Treating trauma within this critical 60-minute window reduces mortality by 50%, and every additional 1-minute reduction in response time improves severe trauma survival by 7% to 10% [2]. However, emergency transit in India is plagued by chronic delays. The national average Emergency Medical Services (EMS) response time is 22 minutes, severely failing the World Health Organization standard of 8 to 10 minutes [2]. In 2024, the state of Odisha recorded 6,142 road accident fatalities [1]. Based on a 47.6% preventability index derived from global trauma literature, overcoming these pre-hospital delays targets a preventable mortality pool of approximately 1,754 deaths annually in Odisha alone [1].

Current Emergency Vehicle Preemption (EVP) systems suffer from eight fundamental systemic failures: (1) queue spillback where civilian vehicles physically block the cleared lane; (2) secondary gridlock or "green starvation" penalizing cross-traffic; (3) single-modality dependence leading to failures in urban canyons or poor weather; (4) complete blindness to non-signalized blockages; (5) prohibitive hardware scaling costs reaching $5,000 per intersection; (6) cybersecurity vulnerabilities and spoofing risks in legacy optical systems; (7) "preemption rebound" causing massive queue ripples upon abrupt signal termination; and (8) the lack of a unified ambulance identity across government and private operators [1], [3], [4], [5].

Fig. 1. EMS Response Times: National vs. Global Benchmarks

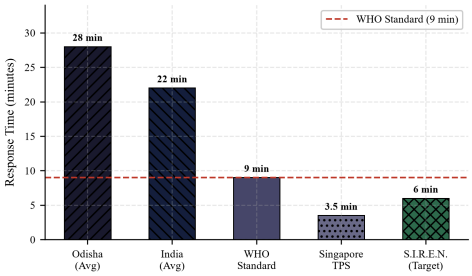


Fig. 1. EMS Response Times: National vs. Global Benchmarks.

Consequently, five critical real-world bottlenecks remain unsolved by conventional intelligent transit frameworks: (1) local lanes lacking traffic lights entirely; (2) disjointed identity fragmentation between 108 and private fleets; (3) civilian driver confusion regarding which side of the road to yield; (4) pedestrian ambiguity near the emergency vehicle; and (5) the absence of mitigation mechanisms for non-traffic jams caused by crowds, disputes, or unauthorized roadside vendors [5]. This paper introduces the S.I.R.E.N. architecture as a comprehensive solution to these five core bottlenecks, providing a robust framework to eliminate pre-hospital transit delays.

II. EXISTING WORK

Several Indian smart city initiatives have attempted to resolve emergency transit blockages. AMBUTAG in Thiruvananthapuram utilizes active RFID, sub-GHz V2X, and GPS on a 2.7 km corridor, achieving a 45–50% travel time reduction. However, it requires hardware tags on all vehicles—rendering it prohibitively expensive to scale—and lacks macro-routing capabilities [2]. B-ATCS in Bengaluru employs YOLO vision cameras and cellular GPS, projecting a >30% transit time reduction for emergency vehicles across 165 signals; unfortunately, it suffers from >500 ms cellular latency spikes and GPS urban canyon multipath errors [2]. The National Organ Transport Green Corridors rely on GPS tracking paired with manual police overrides. While this achieves the highest efficacy at a 60–80% travel time reduction, it is human-intensive, ad-hoc, and provokes severe secondary gridlock [2]. Conversely, the Nagpur IITMS represents a centralized AI approach that projected a 25–35% reduction but experienced catastrophic execution failure, with only 10 out of 171 planned junctions actively functioning [2].

On a global scale, benchmark implementations reveal differing priorities. In Singapore, the Traffic Priority System (TPS) utilizes edge V2X at a 200-metre radius to save 100 seconds per trip, actively preventing gridlock via an ACES whitelisting protocol [3]. Suwon, South Korea implemented a centralized cloud and GPS architecture that successfully reduced emergency vehicle intersection accidents to zero [3]. In the United States, legacy Opticom and Glance systems utilize infrared, cellular, and cloud networks to achieve a 20–31% overall reduction in travel time and a 70% reduction in intersection crashes [3]. However, as demonstrated by a 2025 study in Ghent, Netherlands, isolated signal preemption (saving 27.88 seconds per intersection for emergency vehicles) can severely penalize civilian cross-traffic, increasing delays by up to 47.56 seconds and queuing lengths by 13.44 meters [3].

TABLE I: COMPARISON OF EXISTING EMERGENCY VEHICLE PREEMPTION SYSTEMS

Project	Technology	Scale	Efficacy	Limitations
AMBUTAG (Thiruvananthapuram)	RFID, Sub-GHz V2X, GPS, EVATL	2.7 km (9 signals)	45–50%	Hardware tags on all vehicles; no macro-routing
B-ATCS (Bengaluru)	YOLO, GPS ePath, CoSiCoSt, ASTraM	165+ signals	>30% EV, 15–20% civilian	Cellular latency; GPS urban canyons
Organ Green Corridors (National)	GPS + Manual Police Override	Ad-hoc city-wide	60–80%	Human-intensive; severe secondary gridlock
Nagpur IITMS	Centralized ATCS, ANPR, RLVD	Planned 171 (10 active)	25–35% projected	Execution failure; vendor mismanagement

III. SYSTEM MODEL

The S.I.R.E.N. backend, termed EROS, is designed as a single-process, event-driven FastAPI application structured across seven distinct layers. Layer 1 (Client) comprises a React-based Command Center, a driver mobile application, and a dispatcher console. Layer 2 (Edge/IoT Hardware) encompasses the intersection-level nodes, incorporating YOLOv8 visual obstruction detection and acoustic TDOA siren classification. Layer 3 integrates third-party utilities including OSRM for dynamic turn-by-turn routing and Mapbox GL JS for spatial rendering. Layer 4 acts as the Cloud Backend, orchestrating concurrent tasks such as WebSocket state broadcasting and geospatial logic. Layer 5 (Event/IoT Broker) utilizes HiveMQ Cloud for 1 Hz high-frequency telemetry over TLS and Upstash Redis for an asynchronous write-behind buffer. Layer 6 (Data Storage) stores state using PostgreSQL extended with PostGIS for high-accuracy SRID 4326 calculations. Finally, Layer 7 (RoIP Alerting Gateway) interfaces with police radio channels, synthesizing AWS Polly TTS alerts and dispatching them via a Discord voice bot gateway to simulate live VHF/UHF radio frequencies [7], [8].

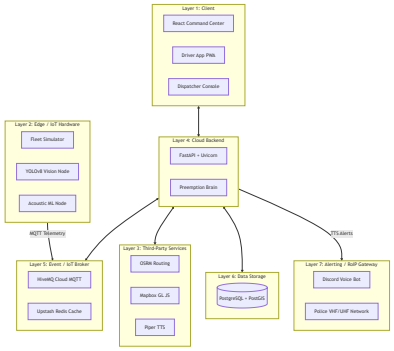


Fig. 2. S.I.R.E.N. 7-Layer End-to-End System Architecture.

To eliminate the systemic vulnerabilities associated with single-modality dependence, S.I.R.E.N. introduces a trilateral sensor fusion architecture. The primary vector relies on Cloud GPS and 5G networks to provide macro-routing and predictive corridor preparation across city-wide ranges at a latency of 100–500 ms. The secondary vector leverages 600 MHz sub-GHz UHF V2X transceivers to ensure deterministic, cloud-independent micro-actuation within a 400-meter intersection radius at under 50 ms latency. Ultimately, the tertiary vector functions as a localized fail-safe utilizing Edge AI; it combines YOLOv8 obstruction detection and acoustic TDOA microphone arrays to operate autonomously within the line of sight at a 55–90 ms latency [2], [3].

IV. HARDWARE

The physical edge node deployed at traffic intersections acts as the localized brain of the S.I.R.E.N. architecture. The core compute is powered by either an NVIDIA Jetson Nano or a Raspberry Pi 5 equipped with a Coral TPU, which runs the YOLOv8 visual inference and audio classification models directly at the junction [4], [6]. For acoustic detection, each node incorporates a sensor array of 3 to 4 INMP441 MEMS microphones coupled with an Analog-to-Digital Converter (ADC) to determine the Time-Difference-of-Arrival (TDOA) of approaching sirens [4], [6]. An HD IP camera integrates with existing CCTV infrastructure or acts as a standalone unit to monitor physical blockages, crowd formations, and ambulance strobe light patterns [4], [6].

At the vehicle level, the edge unit utilizes a u-blox NEO-6M GPS module, 4G/LTE Cat-M cellular connectivity for live location broadcasting, and an optional 600 MHz UHF V2X transceiver for decentralized micro-actuation [4], [6]. Intersection signal overrides are executed physically via a localized relay or RS-485/CAN controller board, which interfaces seamlessly with legacy traffic controllers to initiate priority phases without requiring a complete hardware replacement [4], [6]. To guide civilian traffic, a MAX7219 LED directional arrow matrix dynamically signals which side of the corridor to clear [4], [6]. For overarching resilience during grid outages or cellular dead zones, the nodes run on solar and battery backup power while utilizing a LoRa SX1278 sub-GHz mesh for connectivity fail-safe [4], [6].

TABLE II: HARDWARE BILL OF MATERIALS

Component	Suggested Tech	Purpose
Edge compute node	NVIDIA Jetson Nano/Orin Nano or Raspberry Pi 5 + Coral TPU	YOLOv8 inference + audio classification at junction
Acoustic sensor array	3–4 MEMS microphones (INMP441) per node + ADC	Siren detection & direction-of-arrival
Camera	HD IP camera (existing CCTV + new low-cost nodes)	Obstruction, crowd, ambulance strobe detection
Vehicle unit	GPS module (u-blox NEO-6M) + 4G/LTE Cat-M + optional 600 MHz UHF V2X transceiver	Live location broadcast; junction micro-actuation
Signal interface	Relay/controller board (RS-485 or CAN) to existing signal controller	Priority phase without replacing legacy hardware
Roadside guidance	LED directional arrow board / MAX7219 matrix	Tells civilian drivers which side to clear
Connectivity fail-safe	LoRa SX1278 (sub-GHz mesh)	Keeps nodes talking when cellular fails
Power	Solar panel + battery backup	Resilience during grid outages

V. SOFTWARE

The S.I.R.E.N. software backbone is meticulously structured to separate high-frequency telemetry ingested via MQTT from standard asynchronous REST and WebSocket queries.

A. MQTT Topic Mapping

The system utilizes five dedicated QoS-1 MQTT topics to manage event-driven actuation across the network [7], [8]: - **`sih/ambulance/telemetry/{id}`**: Handles 1 Hz GPS telemetry packets detailing vehicle status, speed, and location. - **`sih/ambulance/route/{id}`**: Distributes OSRM-calculated routes using a GeoJSON LineString format. - **`sih/traffic/signals/{id}/override`**: Commands physical edge nodes with state changes to activate or deactivate the priority phase. - **`sih/police/trigger/{id}`**: Triggers visual and acoustic alarms inside designated police units. - **`sih/roip/channel/{id}/broadcast`**: Pushes dynamic Text-to-Speech instructions to the gateway.

B. Text-to-Speech (TTS) and RoIP Dispatch Pipeline

To bridge digital alerts to traditional police VHF/UHF analog radios (e.g., 136–174 MHz), the architecture is designed around a Radio over IP (RoIP) gateway. However, due to strict telecommunications regulations prohibiting unauthorized civilian broadcasting on official police frequencies, physical radio transmission is legally restricted during the prototyping phase. Consequently, the current S.I.R.E.N. prototype simulates this RoIP integration by routing the synthesized AWS Polly TTS alerts through a lightweight Discord voice bot using `discord.py` and `FFmpegPCMAudio` [6], [7].

C. API Endpoints

The architecture exposes 13 distinct endpoints supporting the client layer [7], [8]:

- **Authentication:** `POST /api/v1/auth/register`, `POST /api/v1/auth/login`
- **Ambulances:** `GET /api/v1/ambulances/active`, `POST /api/v1/ambulances/register`
- **Geospatial Mapping:** `GET /api/v1/map/ambulances`, `GET /api/v1/map/police`, `WS /api/v1/ws/live-map`
- **Emergency Management:** `POST /api/v1/emergency/ambulances/{id}/sos`, `GET /api/v1/emergency/incidents/active`, `PATCH /api/v1/emergency/incidents/{id}/resolve`
- **Manual Overrides:** `POST /api/v1/traffic/override`
- **Routing & Health:** `POST /api/v1/routing/calculate`, `GET /health`

D. Database Schemas

The PostgreSQL storage is structurally divided into six primary tables leveraging PostGIS [7], [8]: - **`ambulances`**: Tracks UUIDs, driver profiles, BLS/ALS equipment tiers, operational statuses, and dynamic WGS-84 location geometries. - **`hospitals`**: Catalogs facility names, care types, and exact geospatial coordinates. - **`traffic_signals`**: Records intersection identities, structural locations, and live control states (AUTO/GREEN/NORMAL). - **`police_units`**: Defines unit call signs (e.g., OD-PCR-001) and designated radio frequency channels. - **`emergency_incidents`**: Maps active crises linking specific ambulances with responding police units based on manual or AI triggers. - **`users`**: Enforces system-wide Role-Based Access Control (Admin/Dispatcher) via bcrypt hashed credentials.

VI. WORKFLOW

The end-to-end execution pipeline of S.I.R.E.N. seamlessly transitions from initial authentication to localized hardware actuation across 15 automated steps [7]: 1. **Auth**: The user authenticates securely, receiving a PyJWT HS256 token.

1. **World State Initialization**: The React Command Center fetches initial ambulance and signal coordinates to render the Mapbox GL JS dashboard.
2. **Route Calculation**: The backend queries OSRM, generating a GeoJSON LineString published via MQTT.
3. **Route Persistence**: The active trajectory is saved as `active_route_geom` within PostGIS.
4. **Live Telemetry**: The in-vehicle IoT unit emits high-frequency GPS telemetry at 2-second intervals.
5. **Write-Behind Redis**: Telemetry packets hit the Upstash Redis buffer for instantaneous O(1) in-memory storage.
6. **WebSocket Broadcast**: The coordinate updates are pushed synchronously to the dashboard via WebSockets.
7. **Preemption Evaluation**: The in-memory Preemption Brain continually assesses spatial rules, checking if an ambulance falls within a 500-meter radius and a 120-degree Field of View (FOV) of a traffic signal.
8. **Signal Override**: Upon matching the criteria, the backend publishes a GREEN status payload over MQTT to the respective edge node for immediate intersection actuation.
9. **TTS Dispatch**: AWS Polly generates a speech alert corresponding to the override, enqueueing it for RoIP broadcast.
10. **Write-Behind Flush**: Every 5 seconds, an asynchronous event loop flushes the Redis buffer, executing a batch UPSERT of all locations to PostgreSQL.
11. **Auto-Reroute**: If the flushed GPS coordinates deviate from the `active_route_geom` by more than 50 meters, the system automatically triggers a route recalculation via OSRM.
12. **Police Dispatch**: In the event of a reported blockage, a PostGIS nearest-neighbor query locates and alerts the closest police unit.
13. **ML Failsafe**: The localized YOLOv8 vision node acts autonomously, detecting physical obstructions and triggering SOS dispatches without manual intervention.
14. **Signal Restore**: Once the ambulance exits the 500-meter radius of the actuated signal, the backend reverts the state to NORMAL, extinguishing the priority phase and mitigating preemption rebound.

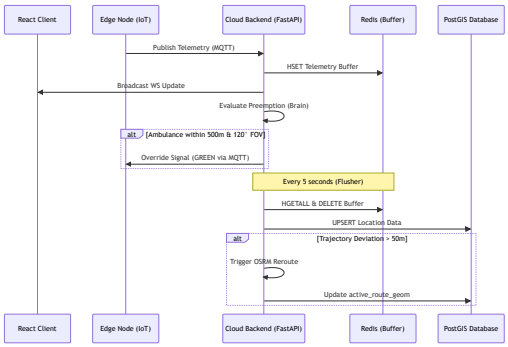


Fig. 3. End-to-End Telemetry and Preemption Sequence Diagram.

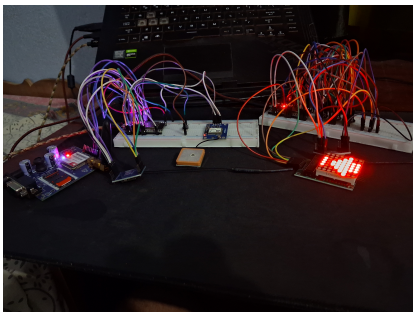


Fig. 4. S.I.R.E.N. IoT Hardware Edge Node Prototype with Sub-GHz Transmitter.

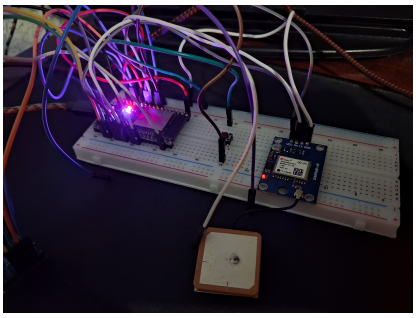


Fig. 5. Dual Edge Node Setup Demonstrating End-to-End V2X Communication.

VII. ALGORITHM

This section details the mathematical models and logic underpinning the S.I.R.E.N. architecture, ranging from spatial calculations to Deep Reinforcement Learning (DRL) and acoustic processing.

A. Acoustic TDOA Multilateration

To determine the approach vector of an emergency vehicle autonomously, the edge node employs a Time-Difference-of-Arrival (TDOA) algorithm via a MEMS microphone array. The time delay τ_{ij} between any two microphones i and j is estimated by maximizing their cross-correlation function. To sharpen the correlation peak in reverberant urban environments, the Generalized Cross-Correlation with Phase Transform (GCC-PHAT) is applied [9]. The mathematical expression is:

$$\tau_{ij} = \arg\max[R_{ij}(\tau)]$$

Once the delays are estimated, the spatial coordinates of the ambulance are resolved using Linear Least Squares (LSQR) multilateration by minimizing the error between measured and predicted delays [9]:

$$\min \sum (\tau_{\text{measured},ij} - \tau_{\text{predicted},ij})^2$$

B. GPS Kinematics and Preemption Brain

The centralized "Preemption Brain" continually evaluates the active fleet's telemetry. The distance between the vehicle and a traffic signal is calculated using the Haversine formula [7]:

$$a = \sin^2\left(\frac{\Delta lat}{2}\right) + \cos(lat_1)\cos(lat_2)\sin^2\left(\frac{\Delta lng}{2}\right)$$

$$d = 2R \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

Simultaneously, the vehicle's heading is calculated to determine if the signal falls within the 120° forward Field of View (FOV) [7]:

$$x = \sin(\Delta lng)\cos(lat_2)$$

$$y = \cos(lat_1)\sin(lat_2) - \sin(lat_1)\cos(lat_2)\cos(\Delta lng)$$

$$\theta = (\text{degrees}(\text{atan2}(x, y)) + 360) \pmod{360}$$

The core Preemption Brain operates via the following event-driven pseudocode:

Algorithm 1: Dynamic Preemption Control Brain

Input: Ambulance identifier a, GPS coordinates (ϕ, λ), operational status s

Output: MQTT override command to traffic signal edge node

```
if s ∉ {active, blocked, dispatched} then return
if a ∈ PREV_LOCS then
    θ_v ← ComputeBearing(PREV_LOCS[a], (φ, λ))
else
    θ_v ← undefined
end if
PREV_LOCS[a] ← (φ, λ)
for each signal node σ ∈ CACHED_SIGNALS do
    d ← HaversineDistance((φ,λ), (φ_σ,λ_σ)) × 1000  {metres}
    key ← (a, σ.id)
    if θ_v ≠ undefined then
        θ_s ← ComputeBearing((φ,λ), (φ_σ,λ_σ))
        Δθ ← |θ_v − θ_s|
        if Δθ > 180 then Δθ ← (360 − Δθ)
        in_front ← (Δθ ≤ 60°)
    else
        in_front ← true
    end if
    if d ≤ 500 and in_front then
        t_now ← CurrentTimestamp()
        if (t_now − COOLDOWNS[key]) > 15 s then
            ACTIVE_OVERRIDES ← ACTIVE_OVERRIDES ∪ {key}
            COOLDOWNS[key] ← t_now
            Publish(σ.id, status ← GREEN)
            Dispatch TTS alert for ambulance a at signal σ
        end if
    else
        if key ∈ ACTIVE_OVERRIDES then
            ACTIVE_OVERRIDES ← ACTIVE_OVERRIDES \ {key}
            Publish(σ.id, status ← NORMAL)
        end if
    end if
end for
```

C. Cross-Traffic Rebalancing and Signal Transitions

To mitigate secondary gridlock, cross-traffic phases are dynamically rebalanced. The baseline cycle length is modeled using Webster's formula:

$$C_0 = \frac{1.5L + 5}{1 - \sum y_i}$$

where L represents lost time per cycle and $\sum y_i$ is the sum of critical flow ratios [9]. To optimize this dynamically, a Deep Reinforcement Learning (DRL) Markov Decision Process (MDP) is introduced. The state space encompasses queue lengths and waiting times, while the action space dictates the dynamic extension of green durations. The objective function minimizes cumulative urban network waiting time while penalizing queue spillback [9].

TABLE III: SIGNAL TRANSITION ALGORITHM COMPARISON

Algorithm	Mechanism	Impact on Network	Recovery Speed
Dwell Transition	Holds green indefinitely	Severe; severs coordination	Very slow; massive queue
Add Only Transition	Caps adjustment (e.g., 20 s/cycle)	Moderate; controlled resync	Moderate; multiple cycles
Smooth Transition	Redistributes lost time proportionally	Minimal; diffuses delay	Fast; statistically proven best
Green Flush	Reverse-sequence downstream clearing	High corridor disruption	Variable; prevents EV entrapment
Logical Quit (LQ)	Autonomous fallback to local actuation	N/A (fail-safe)	Immediate hardware-level

VIII. SPECIFICATION

The architectural specifications dictate strict performance targets and cybersecurity frameworks.

A. Performance Targets and I/O Constraints

The backend operates as a single-process ASGI application. Telemetry arrives at a 1-2 second publish rate via MQTT (QoS 1, TLS port 8883) in a standardized JSON payload mapping WGS-84 coordinates [7]. The processing targets are highly constrained: - MQTT receive to WebSocket push (hot path): < 100 ms - Preemption evaluation (in-memory): < 50 ms per tick - Redis HSET write: < 10 ms - Redis event to PostgreSQL UPSERT debounce: ≤ 5 seconds - PostGIS nearest police query: < 5 ms (utilizing GIST indexes) - AWS Polly TTS synthesis: 500–1,500 ms [7]

B. Geospatial Thresholds

Five primary geospatial thresholds dictate the system's physics [7], [8]: 1. **Preemption Radius**: 500 meters (maximum approach distance for signal override). 2. **Forward FOV Cone**: 120° (±60° from heading) to prevent triggering signals behind the vehicle. 3. **Cooldown Period**: 15 seconds to prevent re-trigger oscillation for the same signal. 4. **Auto-Reroute Deviation**: 50 meters (using `ST_Distance`) to trigger an OSRM recalculation. 5. **SOS Trigger**: 60 seconds of stationary inactivity triggers a manual SOS event.

C. Environmental Configuration (.env)

The runtime environment enforces several critical assumptions, heavily relying on secure connections [8]:

- `DATABASE_URL`: PostgreSQL connection string.
- `MQTT_BROKER_URL`, `MQTT_PORT`, `MQTT_USERNAME`, `MQTT_PASSWORD`: HiveMQ Cloud credentials.
- `SECRET_KEY`: A 32-byte hex token for JWT signing (HS256 algorithm).

D. Cybersecurity Security Model

The system enforces strict security protocols to prevent impersonation or GNSS spoofing, relying on V2X PKI standards (IEEE 1609.2) and Elliptic Curve Cryptography (ECC) processed via FIPS 140-2 Level 3 Hardware Security Modules (HSMs) [11].

TABLE IV: CYBERSECURITY THREAT MODEL AND MITIGATIONS

Attack	Description	Mitigation
Node Impersonation	Attacker spoofs valid EV V2X credentials	V2X PKI (IEEE 1609.2 / ETSI) certificate authentication
Sybil Attack	Single node generates ghost identities to fake traffic density	Spatial-temporal validation + AI-driven IDS
GNSS Spoofing	False satellite signals to alter vehicle location	RF fingerprinting + inertial sensor fusion
Hardware Tampering	Physical extraction of private keys from OBU	FIPS 140-2 Level 3 HSMs
Replay Attack	Re-transmitting valid historical SREM payloads	Secure timestamps + nonce in ECDSA signature

IX. EXPECTATIONS

A. Projected Performance Metrics

Relative to existing frameworks, S.I.R.E.N. projects significant operational enhancements. While domestic deployments like AMBUTAG achieved a 45–50% transit time reduction, they were bottlenecked by hardware tags [2]. Conversely, centralized systems like Nagpur IITMS experienced 0% actual efficacy due to execution failure [2]. Globally, isolated preemption systems like the Dutch iTLC improved EV transit by 27.88 seconds per intersection but severely penalized civilians [3]. S.I.R.E.N. targets a comprehensive transit time reduction from the current 35+ minutes down to under 15 minutes, aggressively approaching the WHO 8-minute standard [5]. The integration of the DRL MDP is projected to enhance EV efficiency by >40% while uniquely reducing negative civilian impact by up to 50% compared to traditional isolated preemption [9]. Furthermore, utilizing Edge IoT nodes instead of overhead gantries yields a 90% cost reduction per intersection [5]. The end-to-end edge detection latency is constrained to 55–90 ms [2].

B. Edge Case Handling Strategies

The architecture is designed with high fault tolerance, categorized across specific edge cases: - **Network Failures**: Cellular dead zones in urban canyons automatically trigger the LoRa sub-GHz radio mesh for unbroken tracking. MQTT broker connection failures are caught via lazy startup, defaulting signals to AUTO while keeping REST APIs active. Database connection exhaustion is circumvented via the Redis 5-second batch flush. Grid power failures are mitigated by solar panels and battery backups [8], [9]. - **Sensor Occlusion**: Camera degradation from fog or rain is overridden by the MEMS acoustic TDOA array, preventing false positives [9]. - **Civilian Ambiguity**: Driver confusion regarding yield directions is resolved via LED directional arrow boards, while pedestrian ambiguity is addressed through 10-second automated acoustic announcements [4]. Non-signalized blockages (e.g., crowds) are uniquely managed by TTS-dispatching the nearest police unit via RoIP [6]. - **Algorithmic State Errors**: Signal oscillation is prevented by strict 15-second cooldowns [7]. Route deviations exceeding 50 meters instantly trigger OSRM recalculations. "Preemption Rebound" is absorbed by the DRL's compensatory green time allocation [9]. Unknown ambulance IDs are silently skipped, and first-frame GPS packets conservatively assume the signal is in the forward FOV. - **Cybersecurity & Administrative**: V2X spoofing attacks are neutralized by IEEE 1609.2 PKI certificates, JWT access tokens, and AI-IDS anomaly detection [11]. The system natively handles missing police registrations non-fatally and mitigates vendor lock-in through NTCIP-compliant hardware agnosticism [2].

Fig. 6. Preemption Actuation Latency: Architecture Comparison

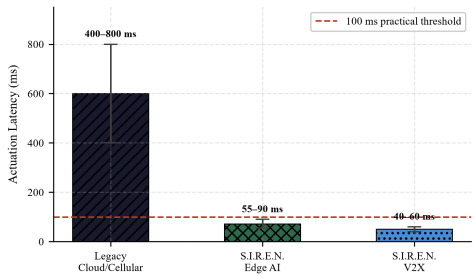


Fig. 6. Preemption Actuation Latency Comparison across Tiers.

X. OUTCOMES

A. Transit Time and Mortality Reduction

By successfully slashing emergency transit times from over 35 minutes to under 15 minutes, the system aligns with critical Golden Hour protocols. Statistically, capturing the 47.6% preventable trauma cohort translates to mitigating a preventable mortality pool of approximately 1,754 deaths annually in the state of Odisha alone [1].

B. SIH Build Roadmap and Local Integration

The deployment strategy follows a rigorous 4-phase Smart India Hackathon (SIH) roadmap:

- **Phase 1 (Weeks 1–2):** Proof of Concept — audio ML siren detection and YOLOv8 obstruction detection, culminating in a basic dashboard.
- **Phase 2 (Weeks 3–4):** Simulation Integration — SUMO testbed validating signal preemption, compensatory green logic, and mock V2X networking.
- **Phase 3 (Weeks 5–6):** Hardware Demonstration — Raspberry Pi/Jetson, microphone array, camera, and LED arrow board integrated into a live physical rig.
- **Phase 4 (Final Week):** Pitch Package — unified API, impact metrics, cost breakdown, and demonstration video [4].

For localized integration in Bhubaneswar, S.I.R.E.N. is designed to overlay onto the 50 existing and 25 newly proposed ATCS junctions. The centralized EROS backend effortlessly plugs into the BSCL ICOMC monitoring center, augmenting manual mobile app tracking with automated geospatial MQTT triggers and TDOA acoustic sensors [9].

XI. CONCLUSION

The S.I.R.E.N. architecture successfully resolves the five critical bottlenecks plaguing emergency transit in developing nations. By combining Edge AI vision and RoIP TTS dispatch, it mitigates non-traffic jams and local lanes lacking traffic signals. The unified EROS

backend eliminates identity fragmentation between government and private fleets. Finally, the physical integration of LED directional boards and acoustic announcements eradicates civilian and pedestrian ambiguity. Through trilateral sensor fusion, write-behind caching, and DRL cross-traffic rebalancing, S.I.R.E.N. presents a highly scalable, cost-effective framework ready for immediate municipal integration to drastically reduce pre-hospital mortality.

XII. REFERENCES

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